

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION**PHYSICS | SSC-I (9th Grade) | Units 1-6 and 9 | Marking Report**

Exam: exam-11-ch1-6-9 (Version 1) Candidate: Seemab Date: 23 July 2026 Total Marks: 65

Deep-check verified: marked page by page against the answer key, then every Section B and Section C calculation was independently re-derived from scratch (not just letter-matched to the key) – screw gauge least count/diameter, motorbike and free-fall kinematics, momentum/impulse (airbag), moment balance, wheelbarrow and manometer pressure, bullet-recoil and bomb-fragment momentum, and the free-fall energy numerical all recomputed independently and match Seemab's working exactly. Section A checked letter by letter against the key (12/12 confirmed, every MCQ correct). A follow-up verification pass caught one additional incomplete answer missed on the first read: Q2(vii) gives valid examples for Optics and Classical Mechanics but never states one for Quantum Mechanics, though the question asks for "one real-world example of each". Three questions (2(vii), 2(ix), 2(xi)OR) and one sub-part (3(i)OR-c) answer only part of what was asked; see the deduction log below. Final verified score: **60.5/65 (93.1%)**.

60.5 / 65 93.1% **A+**

Section A: 12/12 | Section B: 29.5/33 | Section C: 19/20 | 4.5 marks lost across 4 questions, all from incomplete (not incorrect) answers

SECTION A: MCQS

12/12

100% (perfect)

SECTION B: SHORT

29.5/33

89.4% (3 questions incomplete)

SECTION C: LONG

19/20

95% (1 missing verification step)

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (abbreviated)	Seemab	Correct	Result
1	Best instrument to measure a 2 m curtain rod	C	C (Measuring tape)	CORRECT
2	Scale always reads 2 g too high (calibration fault)	B	B (Systematic error)	CORRECT
3	Regular back-and-forth swing of a pendulum bob	B	B (Vibratory motion)	CORRECT
4	Bus decelerates 18 m/s → rest in 4 s	A	A (4.5 m/s ²)	CORRECT
5	Rocket accelerates upward by expelling gas	C	C (Newton's third law)	CORRECT
6	Momentum of a 15 kg trolley at 4 m/s	A	A (60 kg·m/s)	CORRECT
7	Two people push a stalled car, same direction	B	B (Like parallel forces)	CORRECT
8	Torque: 45 N at 0.6 m perpendicular distance	A	A (27 N·m)	CORRECT
9	Spring extends 10 cm under 40 N load	A	A (400 N/m)	CORRECT
10	Pressure: 150 N on 0.03 m ²	A	A (5000 Pa)	CORRECT
11	Renewable resource harnessing sea-level rise/fall	B	B (Tidal)	CORRECT
12	Statement that can never be tested/proven wrong	D	D (Unfalsifiable)	CORRECT

MCQ score: 12/12 (100%, perfect). Every letter matches the answer key exactly. This is the 3rd consecutive perfect MCQ score across all subjects (after Biology exam-12 and Chemistry exam-16, both 22 Jul 2026) and Physics' first clean MCQ sheet since exam-09 (exam-10 lost Q7 on centre-of-gravity). Physics MCQ record rises to 107/108 (99.07%, 1 error across 9 Physics exams: exam-10).

Section B — Short Answer Questions (11 × 3 = 33 Marks)

All 11 parts attempted. OR variant chosen for 2(ii), 2(iv), 2(xi); Main chosen for the rest.

Q.2(i): Measuring Tape vs Metre Rule & Parallax PrecautionChose: Main **3/3**

Correctly identifies a measuring tape over a metre rule (repositioning a 1 m rule over 12 m introduces cumulative error; the tape is one continuous flexible length). Precaution against parallax stated correctly: eye positioned directly/perpendicularly in line with the scale marking. Full marks.

Q.2(ii): Screw Gauge — Least Count & Wire DiameterChose: OR alternative **3/3**

Least count = pitch ÷ no. of circular divisions = 0.5/25 = **0.02 mm**. Circular scale reading = 15 × 0.02 = 0.3 mm. Diameter = MSR + CSR = 2 mm + 0.3 mm = **2.3 mm**. All correct, matches the key exactly.

Q.2(iii): Motorbike Uniform Acceleration NumericalChose: Main **3/3**

(a) $v = u + at = 9 + (2 \times 5) = \mathbf{19\text{ m/s}}$. (b) $s = ut + \frac{1}{2}at^2 = 45 + 25 = \mathbf{70\text{ m}}$. Both correct.

Q.2(iv): Distance & Displacement of a HikerChose: OR alternative **3/3**

Distance and displacement correctly defined (scalar path length vs. vector shortest straight-line change of position). (a) Total distance = $5+12+5 = 22 \text{ km}$. (b) The 5 km north and 5 km south cancel, leaving net displacement = **12 km, due East**. Both correct.

Q.2(v): Newton's Second Law — Crate NumericalChose: Main **3/3**

Newton's second law stated correctly (acceleration directly proportional to net force, same direction, inversely proportional to mass). (a) $a = F/m = 36/6 = 6 \text{ m/s}^2$. (b) $v = u+at = 0+(6 \times 4) = 24 \text{ m/s}$. Both correct.

Q.2(vi): Momentum & Impulse — AirbagChose: Main **3/3**

Momentum and impulse both correctly defined ($p=mv$; impulse = $F \times t$ = change in momentum). (a) $\Delta p = 70 \times (0-15) = -1050 \text{ kg} \cdot \text{m/s}$. (b) $F = \Delta p / \Delta t = 1050/0.2 = 5250 \text{ N}$. Both correct.

Q.2(vii): Optics, Classical Mechanics & Quantum MechanicsChose: Main **2.5/3**

Optics and Classical Mechanics both correctly defined with valid real-world examples: Optics (light/nature and properties of light, **eyeglasses**), Classical Mechanics (motion of macroscopic objects under forces, **trajectory of a thrown ball**). Quantum Mechanics is correctly defined (mathematical description of microscopic particles' motion/behaviour under force), but **no real-world example is given for it anywhere in the answer**, even though the question explicitly asks for "one real-world example of each" field. **-0.5**: missing Quantum Mechanics example.

Q.2(viii): Like & Unlike Parallel ForcesChose: Main **3/3**

Like parallel forces correctly defined (parallel, same direction; two people pulling a stalled cart from behind). Unlike parallel forces correctly defined (parallel, opposite directions; turning a bicycle's handlebars). Both with valid everyday examples. Full marks.

Q.2(ix): Pascal's Principle & Hydraulic Wheelchair LiftChose: Main **1/3**

Pascal's principle stated correctly (pressure applied to an enclosed fluid is transmitted equally and undiminished to every point of the fluid and to the walls of its container) — 1 mark. **-2**: the question's second half ("How does it apply to a hydraulic wheelchair lift raised by a small pedal force?") was never answered – no mention of the small/large piston pair, or how the same pressure produces a much larger force on the wider piston ($P=F/A$). The answer stops immediately after the principle statement.

Q.2(x): Pressure Definition & Wheelbarrow NumericalChose: Main **3/3**

Pressure correctly defined as force acting perpendicularly per unit area, $P=F/A$, SI unit pascal (Pa) = 1 N/m^2 . $P = 900/0.006 = 150,000 \text{ Pa} = 1.5 \times 10^5 \text{ Pa}$. Correct.

Q.2(xi): Geothermal EnergyChose: OR alternative **2/3**

Correctly explains how geothermal energy generates electricity: heat stored in the Earth's crust, wells drilled to hot underground reservoirs, heat converts water to steam, steam drives turbines connected to generators – matches the key's 2-mark explanation in full. **-1**: the question's second requirement, "state one advantage over fossil fuels" (e.g. abundant, reliable/continuous, no greenhouse gas emissions), was never written.

Section B total: 29.5 / 33 (89.4%) - 8 of 11 questions fully correct; 3 questions lost marks for answering only part of a multi-part instruction

Section C — Long Answer Questions (4 × 5 = 20 Marks)

All 4 long questions attempted, every sub-part completed. OR variant chosen for 3(i), 3(ii), 3(iii); Main chosen for 3(iv).

Q.3(i): Three Equations of Motion & Ball Thrown UpwardChose: OR alternative **4/5**

(a) All three equations of motion correctly listed with a symbols table (u, v, t, a all defined) – 2/2. (b) $t = 18/10 = 1.8 \text{ s} - 1/1$ (written with the unit "m/s" instead of "s", a labelling slip only; the numeric value is correct so no mark is affected). (c) Maximum height $h = (18 \times 1.8) - \frac{1}{2}(10)(1.8)^2 = 32.4 - 16.2 = 16.2 \text{ m}$, correct value. **-1**: the question explicitly asks to "verify using $v^2=u^2-2gh$ "; this check ($0 = 18^2 - 2(10)(16.2) = 324 - 324 = 0$ ✓) was never shown, so only 1 of the 2 marks allocated to part (c) is awarded.

Q.3(ii): Momentum Conservation — Bomb FragmentsChose: OR alternative **5/5**

(a) Correctly explains that the total momentum of the fragments after an explosion must equal zero, so they fly apart in exactly opposite directions with equal-magnitude momenta – 1/1. (b) $4v_1 + 6v_2 = 0 \Rightarrow v_2 = -10 \text{ m/s} - 2/2$. (c) $KE = \frac{1}{2}(4)(15)^2 + \frac{1}{2}(6)(10)^2 = 450 + 300 = 750 \text{ J} - 2/2$. All correct.

Q.3(iii): Manometer — Pressure Difference & Absolute Gas PressureChose: OR alternative **5/5**

(a) Manometer correctly defined as a U-shaped tube partially filled with liquid, used to measure gas pressure; correctly explains $P_{\text{diff}} = \rho gh$ at equilibrium – 2/2. (b)(i) $P_{\text{diff}} = 950 \times 10 \times 0.32 = 3040 \text{ Pa}$ – (ii) absolute gas pressure = $100,000 + 3040 = 103,040 \text{ Pa} - 3/3$. Both correct.

Q.3(iv): Conservation of Energy — Falling ObjectChose: Main **5/5**

(a) Law of conservation of energy stated correctly (energy neither created nor destroyed, only converted; total energy of an isolated system remains constant) – 1/1. (b)(i) $PE \text{ at top} = mgh = 6 \times 10 \times 20 = 1200 \text{ J} - 2/2$. (ii) $KE = PE = 1200 \text{ J} \Rightarrow v^2 = 2(1200)/6 = 400 \Rightarrow v = 20 \text{ m/s} - 2/2$. All correct.

Section C total: 19 / 20 (95%) - 3 of 4 long questions fully correct; 1 mark lost for an omitted verification step

Complete Deduction Log (reconciles to headline score)

Question	Awarded	Lost	Reason
Q.2(vii) Main — Optics/Classical/Quantum Mechanics	2.5/3	−0.5	Quantum Mechanics correctly defined, but no real-world example was given for it, though the question explicitly asks for "one real-world example of each" field.
Q.2(ix) Main — Pascal's Principle	1/3	−2	Principle stated correctly, but the required application to a hydraulic wheelchair lift (small piston/pedal force amplified to a large piston force via $P=F/A$) was never written.
Q.2(xi) OR — Geothermal Energy	2/3	−1	Generation process explained correctly and in full, but the required "one advantage over fossil fuels" statement was never written.
Q.3(i) OR(c) — Max Height Verification	1/2	−1	Correct numeric height (16.2 m), but the explicitly-requested verification step using $v^2=u^2-2gh$ was never shown.
Total deductions		−4.5	65 − 4.5 = 60.5/65 confirmed. Every other Section B and Section C question scored full marks - all four deductions come from incomplete (not incorrect) answers: three stop after giving only part of a multi-part instruction (one missing example of three, one missing application, one missing advantage), and one skips an explicitly requested verification step.

Strengths in this attempt

- **Perfect Section A (12/12)** – every MCQ letter matches the key, the 3rd consecutive perfect MCQ score across all subjects and Physics' first clean MCQ sheet since exam-09.
- Strong grasp of core numericals: screw gauge least count/diameter, motorbike and free-fall kinematics, Newton's second law, momentum & impulse (airbag), bomb-fragment momentum conservation, wheelbarrow pressure, manometer pressure, and free-fall energy conversion – all fully correct with clean, complete working.
- Section C nearly flawless (19/20, 95%) – 3 of 4 long questions scored full marks, including a full first-principles listing of the three equations of motion with correctly defined symbols.
- Distance/displacement and like/unlike parallel forces definitions both stated with the correct scalar-vs-vector and same-direction-vs-opposite-direction distinctions, plus valid everyday examples.

Areas to revise

- **Multi-part "define/state X and give an example/apply Y" questions (−3.5 marks combined):** Q2(vii) defined Optics, Classical Mechanics and Quantum Mechanics correctly but gave a real-world example for only two of the three fields, leaving Quantum Mechanics without one; Q2(ix) gave Pascal's principle but never applied it to the hydraulic wheelchair lift asked about; Q2(xi)OR explained geothermal generation but never stated the requested advantage over fossil fuels. This is now three instances of the file-wide "stopping before finishing every sub-part" pattern within this single paper, reappearing in Physics after two clean papers (exam-09, exam-10 were both flagged pattern-free), and the same pattern was also seen in Chemistry exam-16.
- **Skipping the explicit verification step (−1 mark):** Q.3(i)(c) computed the correct maximum height but never showed the $v^2=u^2-2gh$ check the question specifically asked for. Drill fix: whenever a question says "verify using...", treat that as its own required line, not an optional extra, and write it even when the number is already known to be right.
- Minor labelling slip (no mark lost): time was written as "1.8 m/s" instead of "1.8 s" in Q.3(i)(b) – worth a quick habit-check on units when writing kinematics answers quickly.

Physics Exam Comparison

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	7 Apr 2026	63.5/65	97.7%	12/12	32.5/33	19/20	A+
exam-02-ch3-4-9	10 Apr 2026	62/65	95.4%	12/12	30.5/33	19.5/20	A+
exam-03-ch5	18 May 2026	65/65	100%	12/12	33/33	20/20	A+
exam-04-ch6	22 Jun 2026	55.5/65	85.4%	12/12	29/33	14.5/20	A
exam-07-ch1-6-9	25 Jun 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+
exam-08-ch1-6-9	2 Jul 2026	61/65	93.8%	12/12	30.5/33	18.5/20	A+
exam-09-ch1-6-9	7 Jul 2026	64.5/65	99.2%	12/12	32.5/33	20/20	A+
exam-10-ch1-6-9	16 Jul 2026	64/65	98.5%	11/12	33/33	20/20	A+
exam-11-ch1-6-9 (this)	23 Jul 2026	60.5/65	93.1%	12/12	29.5/33	19/20	A+

exam-11 (60.5/65, 93.1%) sits mid-pack for Physics: a perfect Section A carries the paper, but Section B (29.5/33) is pulled down by three questions that stopped after only part of a multi-part instruction (Q2(vii), Q2(ix), Q2(xi)OR), and Section C loses its only mark to a skipped verification line (Q3(i)OR-c). Physics' subject average adjusts to **94.8% (554.5/585, 9 exams)**, holding its #1 subject position just ahead of Chemistry (94.5%).

Physics exam-11-ch1-6-9 | Attempt 2026-07-23 | Marked & independently re-verified 2026-07-23 | Federal Board Class 9